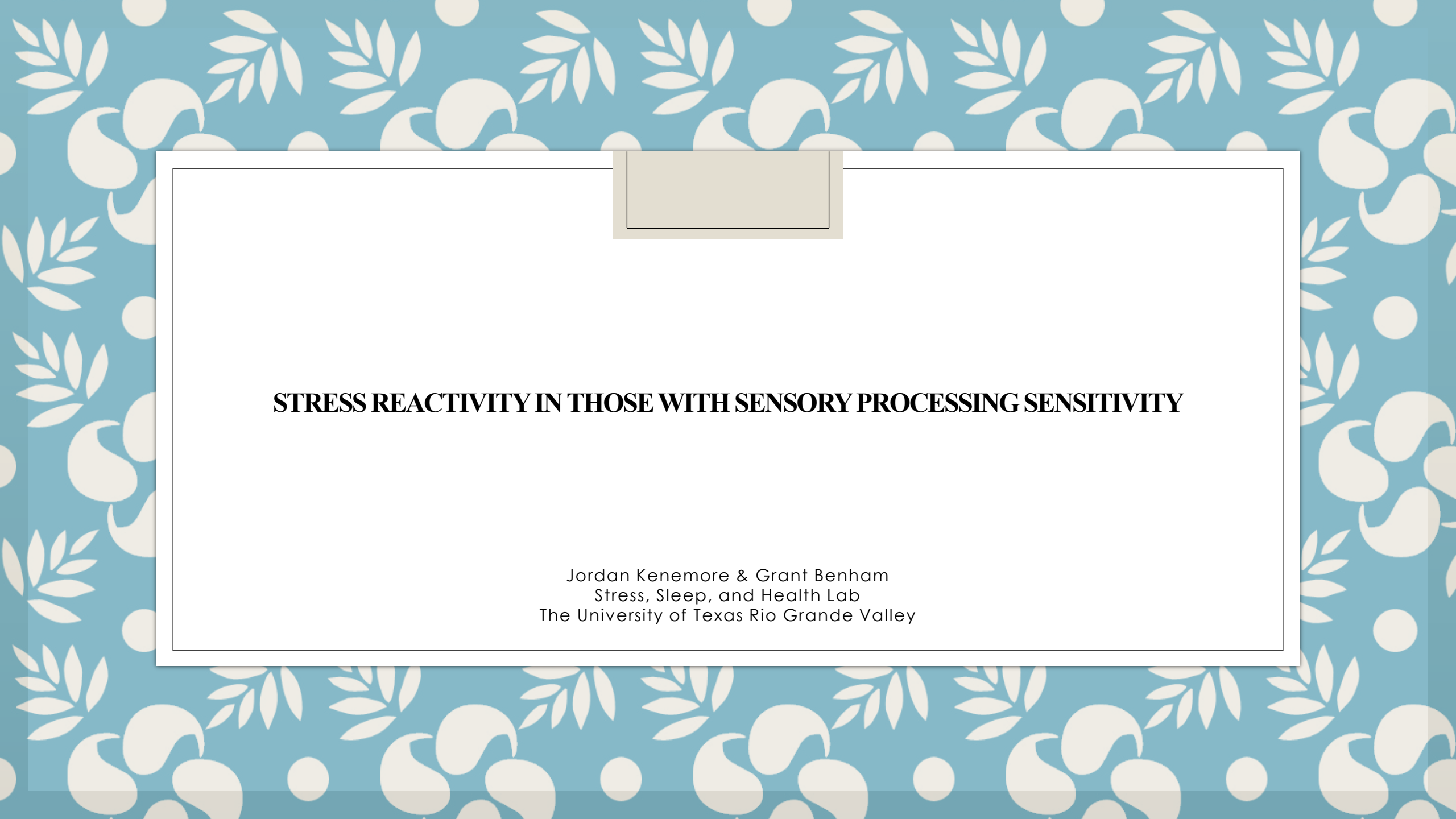




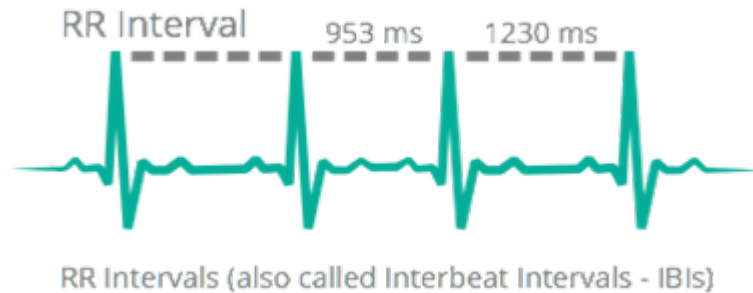
STRESS REACTIVITY IN THOSE WITH SENSORY PROCESSING SENSITIVITY

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- Personality trait
 - Heightened responsiveness to environmental stimuli
 - Intense emotional reactivity
 - Deep cognitive processing^{1, 2}
- Associated with:
 - Poorer mental health (e.g., depression³, anxiety⁴, perceived stress⁵)
 - Poorer physical health (subjective)⁵
- Subfactors:
 - Low Sensory Threshold
 - Sensitivity to stimuli and risk of overstimulation
 - Ease of Excitation
 - Emotional and cognitive overwhelm in demanding environments

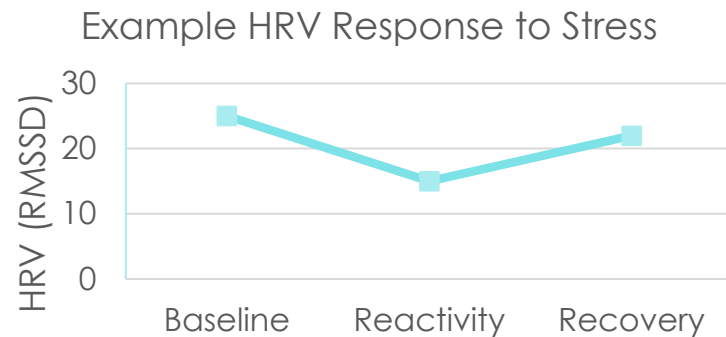
Sensory Processing Sensitivity (SPS)

- Largely under parasympathetic nervous system (PNS) control via vagus nerve
- High HRV – PNS engagement and resilience to stress
- Low HRV – greater stress-related health issues



Heart Rate Variability (HRV)

- Neurovisceral Integration (NVI¹⁵)
 - Higher vagal “tone” indicates better mental and physical health
 - “Resting” HRV serves as indirect measure of self-regulation ability
- Vagal tank¹⁷
 - Builds on NVI, by examining how changes in HRV over time (resting, reactivity, and recovery) are equally important metrics of adaptability
 - *Largely theoretical



Resting HRV vs. Adaptive HRV

- Purpose

- Explore the relationship between SPS and stress reactivity using an objective physiological measure

- Hypotheses

- Negative correlation between SPS and HRV
 - SPS & subfactors will be correlated with stress reactivity during stress task

Current Study

- Sample

- 191 Hispanic (92.7%) females aged 18-30 ($M_{\text{age}} = 19.16$; $SD = 1.29$)

- Measures

- Demographic
 - ECG
 - Modified Montreal Imaging Stress Task (MIST)
 - Highly Sensitive Person Scale (HSPS)
 - Subjective Stress Assessment (Manipulation Check)

Methods

	SPS	LST	EOE
Baseline RMSSD	-0.086	-0.066	-0.127*
Baseline HF-HRV	-0.078	-0.052	-0.094
Stress RMSSD	0.012	-0.005	-0.036
Stress HF-HRV	0.017	.006	-0.015
Reactivity RMSSD	0.154*	0.096	0.144*
Reactivity HF-HRV	0.135*	0.082	0.116

Results

- EOE and resting RMSSD:
 - Heightened autonomic arousal and reduced adaptability to environmental demands
- EOE and HRV reactivity:
 - Greater PNS withdrawal during rest to stress transition
 - Potentially amplify vulnerability to stress-related conditions
- No relationship between LST and HRV:
 - Sensory sensitivity alone may not influence autonomic regulation

Discussion

- Sample limitations
 - Hispanic females
 - Limited age range (18-30)
- MIST (cognitive and social stress test) – limits ability to capture emotionally evocative or sensory specific stressors

Limitations and Future Directions

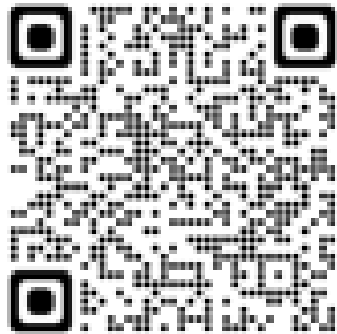
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Thank You